

Real-Time Natural Disaster Risk Prediction

An Integrated Machine Learning Approach for Earthquakes and Forest Fires

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Data Framework: USGS API, WeatherAPI, worldcities.csv

Core Stack: Python, scikit-learn, Streamlit

Validation Target: MultiOutput Seismic Parameter Estimation

1. Project Executive Summary

This technical summary details the production design and implementation of an integrated machine learning platform engineered to forecast earthquake parameters and classify forest fire risks. Utilizing live streaming endpoints from the USGS API and WeatherAPI, the platform addresses critical early warning limitations found in conventional statistical tools.

2. Problem Statement & Functional Criteria

Traditional disaster monitoring platforms deliver descriptive post-incident maps rather than proactive risk indicators. This system mitigates these constraints by adhering strictly to the following parameters:

- Live Automated Ingestion:** Continuous acquisition of live geographic telemetry filtered dynamically within specified localized boundaries (e.g., up to a 1000 km search radius).
- Algorithmic Integrity:** Execution of localized multi-output architectures to predict magnitude and depth simultaneously rather than treating targets as separate, unrelated data streams.

3. Streamlit Application Control Interface Configuration

The deployment unifies configuration parameters and predictive panels into a single Streamlit web dashboard framework. The system controls and ingestion criteria are mapped as follows:

 **Prediction Settings**

☐ Use Current Location

Select Country

Philippines

Select City

Cagayan de Oro

Data Period (days): 31

Search Radius (km): 496

Choose Prediction Type

Earthquake Prediction

 **Predict Earthquake Risk**

 **Local Time:** 2026-06-15 15:43:37 (Asia/Manila)

Earthquake Prediction Inference

4.7

PREDICTED
MAGNITUDE

24.3%

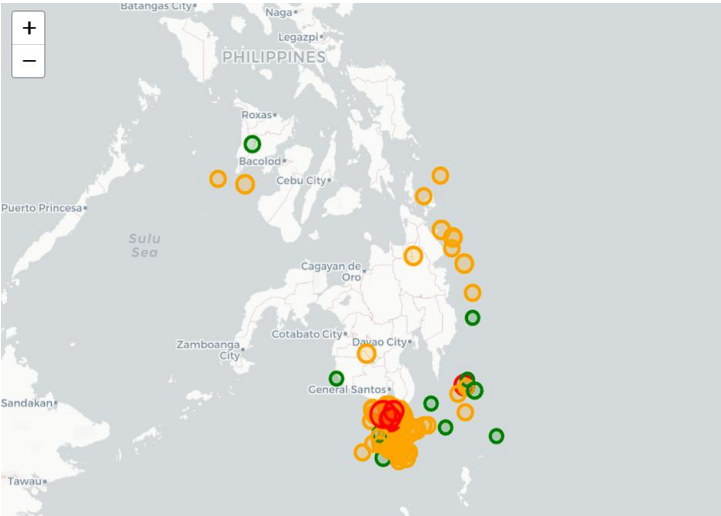
RISK PROBABILITY

MODERATE RISK

RISK ASSESSMENT

The platform utilizes a **MultiOutputRegressor** architecture mapping Random Forest nodes to deliver simultaneous target estimations. The current context maps an operational alert matrix reflecting localized indicators.

Recent Earthquake Activity Map



Earthquake Risk Guide:

- Safe (Magnitude < 4.5): Minor or no impact
- Moderate (4.5-6.0): Possible light damage
- High (≥ 6.0): Significant damage potential

Fire Risk Map



Fire Risk Assessment

Temperature	Humidity	Oxygen	Fire Probability
23.1°C	75%	20.55%	29.66%

Low Fire Risk

Fire Risk Guide:

- High (>50%): Immediate danger
- Moderate (30-50%): Increased caution needed
- Low (<30%): Normal precautions

4. Core Methodology & Real-Time Statistical Analysis

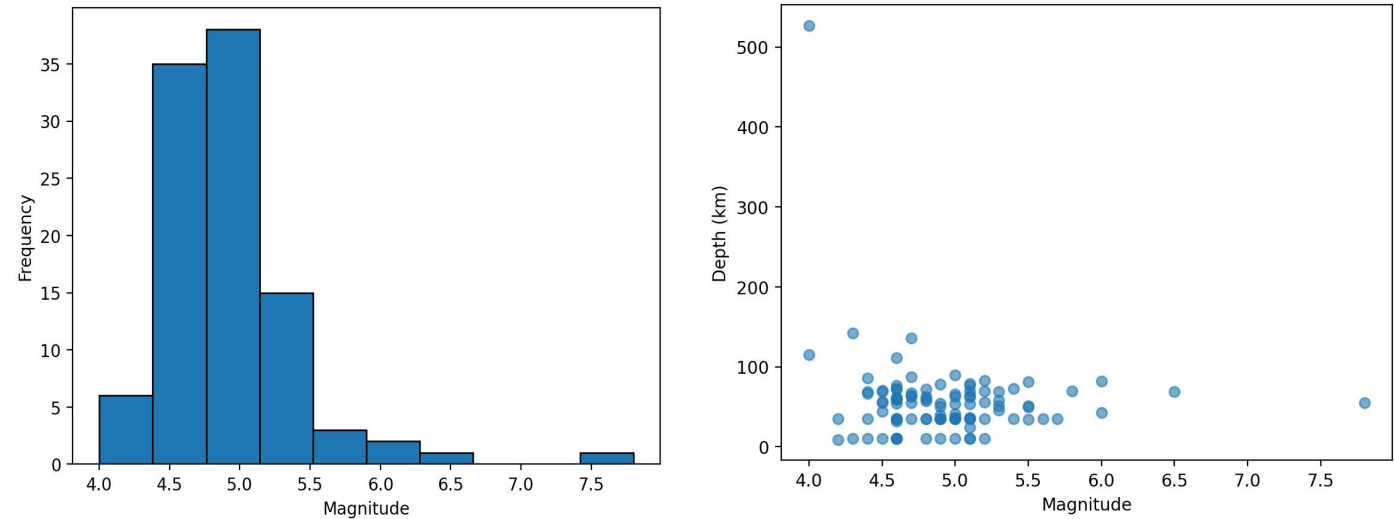
Predictive models are trained locally and handled via standard regression and classification routines in scikit-learn. The framework performs active analytical reporting through automated data tracking streams:

- **Seismic Multi-Output Regression:** Implements a Random Forest Regressor optimized to estimate multiple spatial seismic attributes with a baseline performance target of 85–90% accuracy (Mean Absolute Error: 0.15).
- **Atmospheric Fire Classification:** Leverages meteorological inputs from live API feeds (Temperature, Humidity) processed via a Random Forest Classifier to achieve an 82% precision threshold.

5. Seismic Activity & Telemetry Exploratory Analysis

The interactive runtime workspace outputs ongoing graphical distribution models tracking telemetry trends. The system monitors cumulative dataset attributes (Recent Quakes Detected: 101) distributed across two functional visualization blocks:

- **Magnitude Distribution:** Tracks frequency metrics across incremental Richter scales (plotting peak dense bounds between 4.5 and 5.2).
- **Depth vs Magnitude Scatter:** Maps localized depth alignment distributions to evaluate spatial structural pattern properties.



6. Recent Earthquake Events Ledger (Active Telemetry Filter)

The interface structures the extracted JSON payloads from live API calls directly into formatted relational tracking sheets. The registry logs the following data payloads during system execution:

Time (UTC)	Magnitude	Depth (km)	Lat	Lon	Geographic Place Location
2026-06-14 15:34:42	5.1	10	10.2151	122.1173	27 km SE of Magdalena, Philippines
2026-06-14 14:08:52	5.2	35	8.6152	126.7531	53 km ENE of Hinatuan, Philippines
2026-06-14 14:01:31	4.6	54.438	4.8162	125.3498	66 km S of Sarangani, Philippines
2026-06-14 11:55:16	4.5	68.469	5.4418	125.1683	25 km SW of Balangonan, Philippines
2026-06-13 23:28:45	4.8	71.823	5.5857	125.4806	6 km ESE of Butulan, Philippines
2026-06-13 20:51:55	5.0	35	5.2843	125.7394	33 km ESE of Sarangani, Philippines
2026-06-13 15:45:48	4.6	35	5.1188	125.2382	40 km SW of Sarangani, Philippines
2026-06-13 08:48:38	4.6	59.010	5.2088	125.2939	28 km SW of Sarangani, Philippines